

Hydraulic System

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Designation

BA-Number

Technical Data

Hydropneumatic Spring Assembly System with
Hydraulic Cabinet HSLM for the Master Rollers (M)

BA 911283-01-4

Hydraulic Positioning Device with
Hydraulic Cabinet HSSW for the Slave Rollers (S)

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BA 907011-06-4

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Hydropneumatic Spring Assembly System for the Master Rollers (M)

Technical Data

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1. General

The hydropneumatic spring assembly system is essentially composed of the following groups:

M-spring assembly of mill, complete
M-Roller rotation monitoring, complete
Mill ring line/connecting pipeline, complete
Hydraulic cabinet HSLM.

2. Identification

2.1 Identification of LOESCHE Mill Assembly Groups

Please refer to the parts lists for LOESCHE Mill.

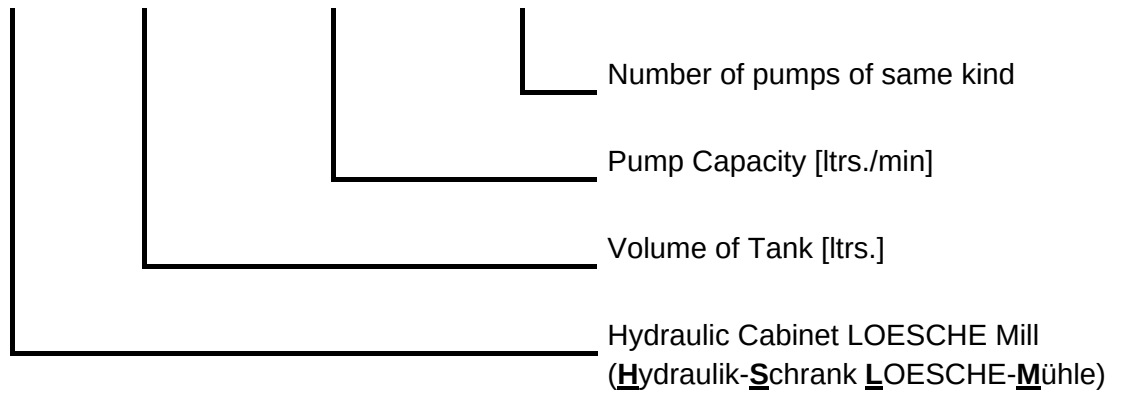
2.2 Identification of Hydraulic Cabinet HSLM

Machine Type	:	HSLM 1000/ ... ¹⁾
Machine No./-Parts List	:	... ¹⁾
Machine Weight	:	... ¹⁾
Machine Drawing	:	see documentation of manufacturer

¹⁾ details : see parts list of hydraulic cabinet HSLM

3. Machine Type Designation of Hydraulic Cabinet HSLM

HSLM 1000 / ... / ...



Hydraulic Positioning Device for the Slave Rollers (S)

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1. General

The hydraulic positioning device for the slave rollers (S) serves the purpose of holding the S-rollers in a preselected distance (working height) in relation to the grinding plate of the table during operation of the mill.

To this effect, the following assembly groups will be essential:

S-spring assembly of mill, complete
S-rotation monitoring of mill, complete
Connecting pipeline of mill, complete
Hydraulic cabinet HSSW.

2. Identification

2.1 Identification of LOESCHE Mill Assembly Groups

Please refer to the parts lists for LOESCHE Mill.

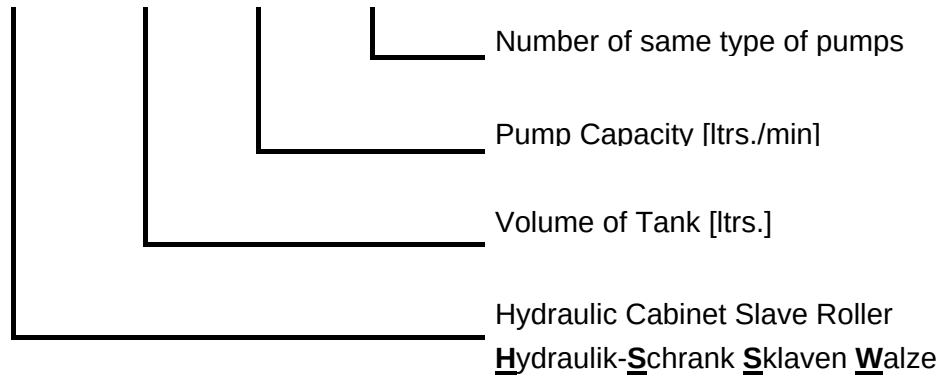
2.2 Identification of Hydraulic Cabinet HSSW

Machine Type	:	HSSW 100/... ¹⁾
Machine No/-Parts List	:	... ¹⁾
Machine Weight	:	... ¹⁾
Machine Drawing	:	see documentation of manufacturer

¹⁾ details: see parts list of hydraulic cabinet HSSW

3. Machine Type Designation of Hydraulic Cabinet HSSW

HSSW 100 / ... / ...



Hydraulic System

Safety

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1. General

These operating instructions Hydraulic System - "Safety" shall be considered as a supplement to the Roller Grinding Mill - "Safety" instructions. It must be ensured that these operating instructions will be studied in detail prior to use of the hydraulic system.

2. Notes on Safety at Work

Particular attention shall be paid to the following notes with regard to safety at work:

- In case of maintenance work (local service of hydraulic cabinets) it must be ensured that no persons will be staying within range of the rollers when the latter are being lifted and/or lowered. Accident hazard!

Distance pieces are to be placed under the raised rollers.
- The limits as specified by the manufacturer for the ratio between operating pressure and filling pressure of hydraulic accumulators shall not be exceeded or fallen short of.
- Maintenance work on the hydraulic system shall not be performed unless the system is depressurized and has been emptied completely.
- Only nitrogen shall be used for refilling of the hydraulic accumulators. Explosions may be caused by the presence of oxygen or air in connection with oil!
- Only the original filling equipment shall be used for filling of the hydraulic accumulators. Attention should be paid to the instructions contained in the hydraulic diagram.
- Filters may not be changed unless it is ensured that the pump motor in the appropriate hydraulic cabinet cannot be started.
- The settings of pressure relief valves (6....) in cabinet HSLM as effected by manufacturer shall not be changed, in order to prevent accidents.
- Quality, cleanliness and operating viscosity of the hydraulic medium are decisive for safety of operation, economy and service life of the machine.

Hydraulic System (M – Roller)

Structure and Function

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1. General

The grinding tools of the LOESCHE Roller Mill (Raw - and Clinker / Slag Machines) are the table and the grinding rollers. The table has a horizontal grinding track which is driven at constant speed through a gearbox. The grinding rollers, which rotate upon a stationary axle without drive, are pressed resiliently upon the material fed onto the table.

Whereas one considers in a raw meal machine that the feedstock will be ground by all the rollers, in a clinker or slag mill one must differentiate between the master and the slave rollers

The rollers (master and slave rollers) are different in size and design depending on the function they serve, and are arranged on the table in opposing pairs. The master rollers comminute the material to be ground by their dead weight and hydraulic pressure additionally applied, and the material will be prepared by the slave rollers merely exerting a light contact pressure.

The pressure of master rollers acting on the material to be ground is generated by the hydropneumatic spring system.

Adjustment of the vertical distance of slave rollers from the grinding path will be effected by means of the hydraulic positioning device.

2. Hydropneumatic Spring System of Rollers

2.1 Structure

The hydropneumatic spring assembly system is essentially composed of the following groups:

2.1.1 Spring assembly, complete

Each roller of the mill possesses a spring unit. This consists, depending upon the type of mill, of one or two hydraulic cylinders that are connected via the rocker arm with the rollers. The hydraulic cylinders are connected in an articulated fashion with the mounting in the mill lower-housing and via the rod head to the rocker arm. The movement of one roller will not have any influence on the movement of the other roller (rollers).

Hydraulic Cylinders

As regards the design of hydraulic cylinders, reference should be made to the appropriate separate documentation provided by the manufacturer. Vent valves are arranged on the cylinders. High pressure hoses connect the cylinders with the fixed connections in the mill stand.

Hydraulic Bladder Accumulators

The accumulator flasks consist of a steel casing with an oil inlet port for connection to the hydraulic pipeline. The inlet port is fitted with a spring-loaded disc valve, which is open during operation. The steel jacket of the accumulator flask encloses a rubber bladder with a gas filler valve, positioned opposite the oil inlet at the upper end of the flask.

Hydraulic Piston Accumulators

These accumulators are similar to a hydraulic cylinder without a piston rod. At one end of the cylinder there is an oil port, at the other end a gas filler valve. A piston equipped with guide-and-sealing rings lies between the oil and gas chambers.

Stop Buffer, complete (for Clinker / Slag Machines)

The stop buffer consists of a welded housing, the adjustable spindle and the biased elastomer for the attenuation of shocks.

The housing will be bolted to the M-stands in the mill lower housing.

Readjustment of the spindle is possible as a function of the wear of milling components. Adjustment will be effected by means of a 10 mm shim plate underneath the M-rollers on the empty table. Herein, the spindle will be screwed in from the outside up to the stop plate of the fork of rocker arm system. A cap will prevent any subsequent twisting of the spindle during operation.

As any indication of wear will be found on the milling components, the spindle adjustment should be checked in due time. Otherwise, the stop buffer will be loaded unnecessarily by hitting of the fork.

2.1.2 Roller Rotation Monitoring, complete ¹

The rotation monitoring of the roller is achieved using a proximity switch and thus avoids metallic contact. The switch is mounted in a pipe which is led through the rocker arm. The impulses are generated by the bolts which attach the slipring sealing to the rotating hub.

¹ only valid for partial load

2.1.3 Ring Line/Connecting Pipework, complete

The oil supply system (hydraulic cabinet) and the hydropneumatic spring system of mill will be interconnected via a pipeline system. The cross-sections for each pipeline as shown in the drawing must be observed. Welded pipes will be descaled by pickling and preserved at the time of manufacture and must be protected from corrosion and the ingress of dirt by hermetic sealing of the flanges until the time of final installation and assembly.

As a rule, the connecting pipeline will be installed in a duct embedded in the concrete foundation of the plant. **This duct must be provided with an oil-resistant paint coating, in order to prevent the penetration of oil into the concrete and the deterioration thereof.** Attention shall be paid to proper drainage of the duct, to permit leakage oil to be drained and to have a possibility of removing dirt by means of water.

2.1.4 Hydraulic cabinet HSLM

The hydraulic cabinet HSLM is the supply unit for the hydropneumatic roller spring system of the mill.

Speciflicated instructions for the design of the hydraulic cabinet HSLM are to be taken from the manufacturer's documentation.

2.2 Mode of Operation

During mill operation the rollers are raised by the material to be ground on the table as a function, as a function of layer thickness. The rocker arm rotates slightly and the pistons of the hydraulic cylinder move up, forcing the oil on the piston rod side out of the cylinder chamber through hoses and piping into the hydraulic bladder accumulators. The oil flowing into the accumulator flasks compresses the nitrogen-filled accumulator bladder. The non-homogeneity of the grinding bed sets up vertical movement in the rollers, as when a car wheel runs along a rough road. If the rollers run into a "grinding bed trough", the accumulator bladder expands and pushes a corresponding amount of oil from the accumulator housing back into the cylinder. This means that there is a continual exchange of oil between cylinder and accumulator, with a very rapid response. In a similar manner the piston accumulators are connected with the piston chamber of the hydraulic cylinder. When oil is displaced from the cylinder into the accumulator the piston there moves and compresses the nitrogen. Should the roller move upwards, oil is again discharged from the accumulator.

The hydraulic cylinders of opposite rollers are interconnected on the oil side so that the rollers will exert the same force on the grinding bed irrespective of their deflection thus ensuring a uniform loading of the thrust bearing in the mill gearbox.

For starting and stopping the mill, it is possible to lift the rollers hydraulically off the grinding bed. This is effected by directing the oil of the pump from the piston rod side to the piston side chamber of the hydraulic cylinder. Owing to the different resistances in the hydraulic system, the rollers are not lifted simultaneously but one after another. The roller movements upwards are mechanically limited. As soon as all rollers have reached their top limit position, this state is electrically signalled.

With the rollers lifted it is possible to start a mill when filled. When starting a filled mill, the mill motor is loaded to appr. 40 % of the operational torque. Upon lowering the rollers, the remaining grinding stock on the table is then ground quickly. The lowering of the rollers occurs with a time delay with respect to the feeder. This increases the grinding bed depth slowly and the motor torque rises correspondingly to 100 %. With this mode of operation the full load condition will be attained after about 30 seconds.

When the spindle (in Clinker / Slag Machines) is properly adjusted the stop buffers additionally ensure that no metallic contact will be possible between the grinding components, i.e. master roller and grinding plate, when material feed to the mill is interrupted and the M-rollers cannot be lifted in due time. The permissible lowest position of the rollers is electrically monitored.

2.2.1 Spring Characteristics

General

The hydropneumatic spring system with the HSLM hydraulic cabinet permits the roller force to be adapted to the material to be ground and to keep the grinding bed thickness constant within close tolerances.

It consists of two separate systems, the so-called operating system which provides pressure to the rod-side of the cylinder and the counterpressure system which supplies pressure to the piston side. The control system is regulated according to the mill feed rate.

Diagram: Roller spring stiffness

The drawing number of "Diagram: Roller spring stiffness" may be gathered from the documentation parts list of the hydraulic cabinet HSLM.

Diagram: Load per roller

The drawing number of "Diagram: Load per roller" may be gathered from the documentation parts list of the hydraulic cabinet HSLM.

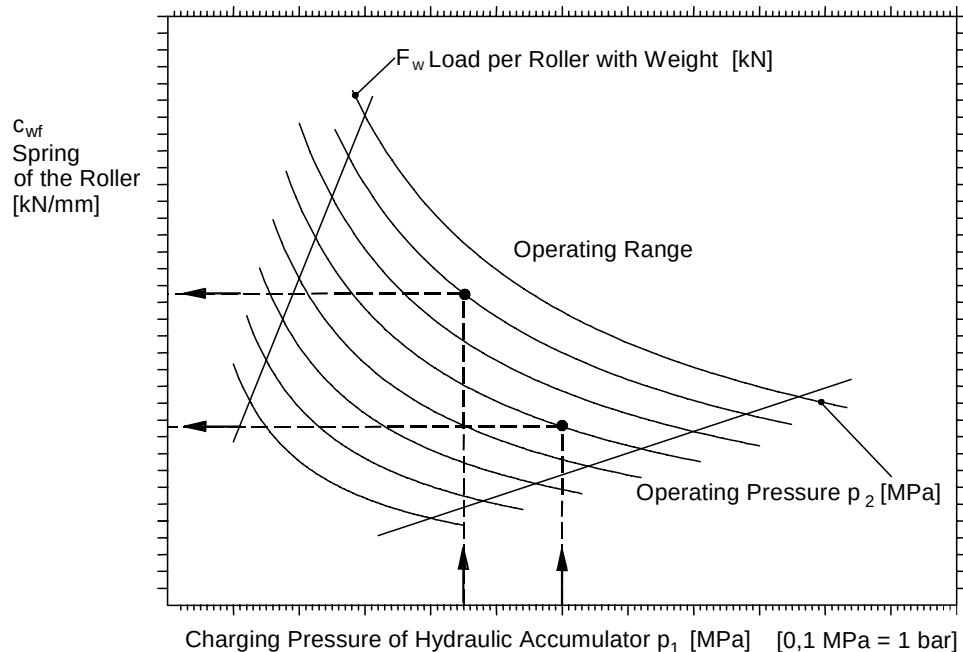
2.2.2 Operating Pressure (Spring Stiffness)

General

The spring characteristic line, i.e. the spring stiffness, can be selected upon the rod side by adjusting the ratio of the hydraulic operating pressure p_2 to the N_2 -charging pressure of the accumulator bladder p_1 and, without variation of the N_2 -pressure, the spring stiffness can even be varied during operation of the mill. The "Diagram: Roller Spring Stiffness" indicates the limits which have to be observed.

Operation

In the "Diagram: Roller Spring Stiffness" the roller stiffness c_{wf} (= spring stiffness related to the roller) is plotted against the nitrogen-charging-bladder pressure p_1 . The parameters are the various hydraulic operating pressures p_2 and the roller forces F_w , respectively.



Determination of spring stiffness of the roller c_{wf}

1. Selection of operating pressure p_2 (with reference to "Diagram: Roller Load" (see documentation parts list "HSLM")).
2. Selection of charging pressure of hydraulic accumulator p_1 within the operating range.

(please note: $p_{1max} = 0.8 * p_{2min}$ $p_{1min} = 0.25 * p_{2max}$)

3. Follow the vertical axis of the accumulator charging pressure p_1 until it intersects the operating pressure curve p_2 .

4. From the intersection follow the horizontal axis left to read the corresponding spring stiffness (c_{wf}) of the roller.

The roller stiffness value is a measure of the amount by which the roller force increases for a roller deflection of 1 mm (0.04 in).

The smaller the difference between p_2 and p_1 , the softer the spring. This means that with a soft spring, the force increase is low when deflecting the roller. The max. p_1 value of the N_2 -charging in the accumulator must not exceed 80 % of the operating pressure p_2 , as otherwise the rubber bladder could be destroyed.

2.2.3 Counterpressure Function

General

Piston accumulators are mounted adjacent to the hydraulic cylinders. This promotes a harmonious transfer of oil (as a result of the roller movements) during mill operation.

Operation

In order to function correctly, the piston accumulators require that a counterpressure be generated on the piston side of the hydraulic cylinder. This counterpressure should be kept low to avoid negative influences upon the rod side operating pressure. It must, however, always exceed 0,7 Mpa (7 bar).

The setting of the counterpressure p_4 is determined in relation to the N_2 -charging pressure p_3 of the piston accumulator.

N. B.: The N_2 -charging pressure p_3 and the counterpressure p_4 of the piston accumulator must be maintained in the range of $\pm 10 \%$ of the values quoted in the "Diagram: Roller spring stiffness".

During mill operation the piston of the accumulator should then lie approximately in the midpoint of its possible travel range (nominal-value).

2.2.4 Roller-Relief-System Pressure (R-R-S) – Partial Load Operation

General

If required, the mill can be operated with a roller relief system.

A prerequisite for this is that a roller rotation monitor is installed. The roller must rotate during mill operation.

Operation

In principle for higher mill feed rates the operating pressure will be controlled and the counterpressure is maintained at a low, constant value. The commissioning engineer adjusts the operating system so that the roller force can be varied from 100 % to approx. 50 %. 100 % operating pressure (i.e. roller force) is the necessary pressure required to achieve the maximum mill product rate and varies with the mineral properties of the feed material. Should, however, a smaller roller force than 50% be required (partial load) then the operating pressure is maintained at its minimum value and the counterpressure is raised. The effective operating pressure (and the roller force) is thereby further reduced.

2.2.5 Functions of the Hydropneumatic Spring System

(For positioning of valves please refer to "Scheme: Hydraulic" of the HSLM)

The pump supplies oil to the hydraulic system. During mill operation it only runs in order to replace oil lost by leakage which leads to pressure loss. The piping in the cabinet is equipped with pressure monitors.

a) Increase (Raising) of Operating Pressure

The required operating pressure and the control characteristics can be pre-selected within the PLC. The pump will automatically start if the said adjustment is performed during mill operation and will stop as soon as the higher value is attained.

b) Reduction (Lowering) of Operating Pressure

The required operating pressure and the control characteristics can be pre-selected within the PLC. Solenoid valve (7.2) will be opened automatically during mill operation so that the oil may return to the tank. The returning oil is led through a throttle (18.2) to control the rate at which the pressure decreases. The solenoid valve (7.2) is closed as soon as the lower pressure has been attained.

Please check on the basis of the Spring Diagram whether the ratio between the N₂-charging pressure and the hydraulic operating pressure is still within the admissible tolerance. If necessary, correct the N₂-charging pressure. Such correction shall not be made unless the mill is at standstill and the operating pressure has been completely relieved.

c) Increase (Raising) of Counterpressure-System

The required pressure and control characteristics can be pre-selected via the PLC. Solenoid valve (7.10) opens and oil flows from the operating system to the counterpressure system. The oil is led through a throttle (18.10) to control the rate at which the pressure increases in the system. The valve shuts as soon as the higher counterpressure has been attained.

d) Reduction (Lowering) of Counterpressure System

The required pressure and control characteristics can be pre-selected within the PLC. Solenoid valve (7.5) opens and allows oil to flow back to the tank. The oil is led through a throttle to control the rate at which the pressure decreases. The valve shuts as soon as the lower pressure has been attained.

e) Roller Raising

With automatic control the solenoid valves (7.1) will be switched into their appropriate position according to hydraulic diagram. With local operation it is instigated by depressing the "Roller Raising" pushbutton.

The pump will start automatically and continue to run until all rollers have reached their uppermost limit position which will be monitored by electrical proximity switches on the rocker arms. Thereafter, the pump will be stopped automatically.

During lifting of the rollers, the oil pressure within the piston rod side hydraulic system will be increased in excess of the set operating pressure. Safety valves (6....) will ensure that the maximum permissible system pressure will not be exceeded.

During lifting of the rollers oil can alternatively be led from the operating pressure side on the counter pressure side. In doing so valve 7.3 is opened temporarily time- or pressure-dependent. A Loesche engineer carries out the calibrations during the commissioning.

f) Retaining of Rollers in their top Positions (Automatic Operation)

Should one or several rollers leave the appropriate uppermost end positions due to leakage within the hydraulic system, the pump will be started automatically in the same way as described under "Roller Raising".

Attention: For safety reasons no persons shall be allowed to stay within range of the movable components of the milling force transmission system.

g) Roller Lowering (Automatic Operation)

Upon opening of the valves (7.5 and 7.6), the oil will be returned from the piston side hydraulic system to the tank, until the pressure has been reduced to that which was preset for the counterpressure system. The lowering speed of rollers will be determined by the LOESCHE commissioning engineer by appropriate adjustment of throttles (18.5 and 18.6).

h) Relieving of Pressure

If the entire hydraulic system is to be relieved to atmospheric pressure, e.g. in case of charging pressure measurement with the accumulators or in case of maintenance work, the system must be switched (from "Remote") to "Local" mode.

In this mode, the control system will be inoperative.

Upon opening of the stopcocks (according to the function table of the hydraulic scheme) by manual operation, the oil will return to the tank.

i) Retaining of Rollers in their top Positions for Service Work

Automatic lifting of rollers.

Clearance for local operation from CCR. Pressure control system is no longer active. The position of the stopcocks with lifted rollers must be according to the table shown in the hydraulic scheme.

Attention: For safety reasons no persons shall be allowed to stay within range of the movable components of milling force transmission system unless additional mechanical safety precautions (e.g. distance pieces between roller and table) will prevent any accidental movement of the rollers.

j) Parallel Flow Filter

When valve 7.1 is in its neutral position and the pump is started, the tank contents are (in addition to the main filtration) circulated over the filter at certain time intervals.

The operational demands of the machine have priority over those of the parallel flow filter system.

Hydraulic System (S – Roller)

Structure and Function

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1. Hydraulic Positioning Device for Slave Rollers (S)

The hydraulic positioning device for the slave rollers (S) serves the purpose of holding the S-rollers in a preselected distance (working height) in relation to the grinding plate of table during operation of the mill.

To this effect, the following assembly groups will be essential:

- S-spring assembly of mill, complete
- S-roller rotation monitoring of mill, complete
- Connecting pipeline of mill, complete
- Hydraulic cabinet HSSW.

1.1 Structure

1.1.1 S-Spring Assembly, complete

Each slave roller of the mill is connected to a hydraulic cylinder supported inside the cover of the mill upper housing. The connections will be pivotable via bolts and self-aligning bearings. A stop buffer is arranged underneath the rocker arm of the slave roller.

Hydraulic Cylinders

Please refer to the appropriate separate documentation provided by manufacturer.

Stop Buffer

The stop Buffer serves to dampen shocks. The major components are the console, the rubber pad and the adjustable spindle.

Adjustment of spindle will be effected by means of a 10 mm shim plate underneath the S-roller on the grinding path. Herein, the spindle will be screwed in from the outside up to the rocker arm of S-roller.

This adjustment will ensure that a metallic contact will not be possible between the S-roller and the grinding plate.

As any indication of wear will be found on the milling components, the spindle adjustment should be checked in due time.

1.1.2 S-Roller rotation monitoring, complete

S-roller rotation will be monitored by means of a shaft introduced through the center bore of roller axle and connected to the rotating roller core cover of S-roller.

A perforated disk with proximity initiator is arranged at the end of the shaft and accessible from the outside. The shaft will be supported within the center bore by means of bushings.

1.1.3 Connecting Pipework, complete

Pipelines are provided for connecting the hydraulic cabinet HSSW with the hydraulic cylinders of S-rollers.

1.1.4 Hydraulic Cabinet HSSW ...

The hydraulic cabinet HSSW is the supply system for the hydraulic positioning device for the S-rollers.

For details as to the design of cabinet HSSW ... reference should be made to the separate documentation provided by manufacturer and the hydraulic scheme HSSW..

1.2 Mode of Operation

During mill operation the slave rollers will be adjusted separately by means of the hydraulic positioning device to a distance above the metallic grinding path permitting an optimal preparation of the grinding bed for subsequent comminution under the master rollers and ensuring that the S-rollers will rotate substantially free from any slip due to friction with the material being ground. Height adjustment will depend on the material to be ground and will be determined at the time of commissioning and initial startup. The height can be read from the stroke monitoring.

Deviations from this working height may be encountered during normal operation (see also hydraulic scheme HSSW ...) under the following conditions:

In case of any non-permissible passage of foreign matter underneath the S-roller or too thick a grinding bed a force will be exerted on the S-roller in upward direction. Should this force exceed the cylinder force resulting from the piston area and the value set on the pressure relief valve, the S-roller will be deflected.

Any oil leakage of the hydraulic system both on the piston and the piston rod side will cause the S-roller to deflect.

1.2.1 Functions of the Hydraulic Positioning Device for the Slave Rollers (S)

For positioning of valves please refer to "Hydraulic Scheme HSSW ..."

(Any information shown in parentheses hereinafter shall apply to S-roller No.2)

a) Raising of Slave Roller No. 1 (No.2)

Upon activation of control valve 40.1B (40.2B) and starting of pump 20.1 (20.2) the S-roller is lifted. The oil displaced from the other cylinder chamber will be returned into the tank via the adjustable throttle 60.3 (60.4), valve 40.1B (40.2B) and filter 16.

b) Retaining of Slave Roller No. 1 (No.2) in top Position

Hold control valve 40.1B (40.2B) in position 1. Pump needs to be started only, when S-roller is being lowered due to oil leakage.

c) Lowering of Slave Roller No.1 (No.2)

Upon activation of control valve 40.1A (40.2A) and starting of pump 20.1 (20.2) the S-roller is lowered. The oil displaced from the lower cylinder chamber will be returned into the tank via the adjustable throttle 60.1 (60.2), valve 40.1A (40.2A) and filter 16.

d) Controlling of Slave Roller No. 1 (No.2) for Service Work

Clearance for local operation from CCR. Pressure control system is no longer active.

Using pushbutton 26.1 (26.2) integrated in hydraulic cabinet HSSW the S-roller may be lifted or lowered.

Attention: For safety reasons no persons shall be allowed to stay within range of the movable components of positioning device.

Hydraulic System

Commissioning/Startup and Operation

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Attention should be paid (among other things) to the following notes for commissioning and initial startup of the hydraulic system:

1. Commissioning and Initial Startup of Hydraulic System

Clean complete hydraulic system with flushing oil.

Fill complete hydraulic system with oil of the specified grade up to the upper level mark in tank - including piping and cylinders.

Fill hydraulic accumulator with nitrogen; the filling pressure will be determined by the LOESCHE commissioning engineer at the time of commissioning and initial startup. The leaded safety valve (6...) in the HSLM hydraulic cabinet needs to be checked to ensure that the seal is not damaged.

Adjustment of the proximity limit switches on the rocker arms as shown in drawing "Rocker Arm Monitoring, complete" (see mill parts list). These proximity switches are provided on each rocker arm of the M-roller.

Checking of the positioning of hand-operated valves within the hydraulic system according to drawings "Hydraulic Scheme HSLM ..." and "Hydraulic Scheme HSSW ...").

Treat the piston rods of the hydraulic cylinders for the S-Roller with rust inhibitor.

1.1 Commissioning and Initial Startup of Hydraulic Positioning Device for the Slave Rollers (S)

At the time of commissioning and initial startup of the mill as well as in case of major changes of the composition of the material to be ground it will be necessary to determine the proper working height for the slave rollers by trial-and-error method. The objective is to minimize the oscillation rate (vibration) of mill to ensure reliable and trouble-free operation of the machine.

To this effect, the following measures are necessary which will be initiated by the LOESCHE commissioning engineer at the time of commissioning and initial startup:

Setting out from the preselected height of slave rollers above the metallic grinding path plane, the S-rollers will be lowered and/or lifted in discrete steps (5 mm to 10 mm). Herein, both the speed of S-rollers and the relative oscillation rate of mill (checked at the mill top section) will be measured simultaneously and recorded in tables in relation to the particular working height involved.

Both S-rollers may be positioned separately. Reasons are the different positive coupling to the grinding bed ensuring rotation of the S-rollers and the difference, if any, in roller diameter due to wear. During the above measurements, however, both S-rollers should move in the same direction (UP/DOWN) and be substantially synchronized.

Using the table so prepared, the working height of S-rollers may be selected as a setpoint for automatic operation at which the oscillation rate of mill is minimum.

The following settings should be made and/or checked for the various pressure relief valves in the HSSW control cabinet:

Pressure relief valves:

Item	50.1	10 MPa
	50.2	10 MPa
	50.3	20 MPa
	50.4	20 MPa
	50.5	16 MPa
	50.6	16 MPa

The settings of throttle valves should be effected at the time of commissioning and initial startup, since they depend on the pipeline routing and the resultant line resistances within the plant concerned:

Items 60.1 and 60.2 have an influence on the lowering speed of slave rollers, and Items 60.3 and 60.4 on the lifting speed. Herein, the setting of throttle valves should be selected so that pressure relief valve 50.1 and/or 50.2 will not open during normal lifting.

2. Starting of Mill in Empty or Filled Condition

After the electric release of hydropneumatic spring system for the master rollers, the following functions will be initiated automatically according to the electrical control sequence. The appropriate positions of hand-operated and solenoid valves may be gathered from the table of function in drawing "Hydraulic Scheme HSLM ...":

- "Working pressure (i.e. pressure increase)"
- "Raising of master rollers":
- "Keep master rollers in top position"
- "Lowering of master rollers"
- "Mill in operation"

As a function of the different resistances within the hydraulic system, the first action will be the lifting of one M-roller/rocker arm unit until a mechanical stop is reached (the stop is given by the limitation of the piston travel at the cylinder head).

After the rollers have reached their upper position, the pump motor is switched off by the proximity switches.

NOTE : While the pistons move out during the lifting of the rollers, the oil of the piston rod chambers is displaced into the hydraulic accumulators and the piston accumulators; as the nitrogen filling is not released, the pressure in both systems is increased relative to the set pressures for the mill operation, when valve 7.3 is not active.

The control system does not function whilst the rollers are being lifted or lowered. The higher oil pressure (piston rod side) can be read from the pressure gauge (4.2), whilst the oil pressure which is used for the lifting process in the lower (piston side) cylinder chamber is indicated on the pressure gauge (4.3).

When the rollers have been lifted and due to oil leakage, one or more of the rollers are moving downwards, the high pressure pump which belongs to that specific roller is automatically started until the switching point "Rollers in top position" is reached (valve 7.1 is activated).

During the procedure "Lowering Rollers" the rollers move downwards onto the table because of gravity and additional force of the gas filling in the accumulators. Whilst the rollers are being lowered the controllers do not function.

For smooth lowering of the rollers onto the grinding bed the return oil from the piston side cylinder chambers is initially fed back into the tank through the adjustable throttle.

The prerequisites to be satisfied for activation of the individual functions may be gathered from the sequence chart forming part of the LOESCHE electrical documentation.

In parallel to the M-rollers, the functions of hydraulic positioning device for the slave rollers will also be initiated upon electric release. The appropriate positions of hand-operated and solenoid valves may be gathered from the table in drawing Hydraulic Diagram HSSW ...":

- "Raising of slave rollers"
- "Keep slave rollers in top position"
- "Lowering of slave rollers" (until correct working height has been attained)
- "Mill in operation".

The prerequisites to be satisfied for activation of the individual functions may be gathered from the sequence chart forming part of the LOESCHE electrical documentation.

3. Shutdown of Mill in Case of Normal Operation

When the mill is to be stopped, the conveyor for the material to be ground will be disconnected. The mill motor may be disconnected with time delay, after the material to be ground and still on the table will have been removed by centrifugal force after several revolutions.

Upon interruption of material feeding, changeover of the hydraulic pressure to the piston side of the spring cylinders will normally be effected within the hydropneumatic spring system so as to lift off the master rollers from the table.

The system reacts correspondingly for lifting this S-rollers.

4. Emergency Stop

In case of an Emergency Stop or a power supply failure both the master rollers and the slave rollers will remain in contact with the material to be ground. Braking of the table will take one revolution maximum.

5. Local Operation

For maintenance work the system must be switched to "Local Operation" in the control room. The normal control system is then inoperative.

The rollers may then be lifted via the hydraulic cabinets HSLM and HSSW in that the pumps will be started locally.

Hydraulic System

Maintenance/Inspection

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1. Cleanliness

Cleanliness of the plant is of decisive importance for proper functioning and a long service life of the components.

2. Hydraulic Oil

Use only the grade of hydraulic oil recommended by LOESCHE. The class of purity of the employed oil at least should conform to **NAS 1638 Class 8 or better**.

Maintain the specified oil change intervals.

3. Maintenance Work

The piston rods of the hydraulic cylinders for the S-rollers are to be regularly checked for signs of corrosion and treated with rust inhibitor if necessary.

With a view to the prevention of accidents no maintenance work shall be performed on the hydraulic system unless the oil pressure is relieved prior thereto.

Unions and screw connections shall be checked for proper seating at regular intervals.

Within the high-pressure area materials may be used only satisfying the requirements for the 21 MPa pressure stage.

4. Filling of Hydraulic Accumulators

4.1 Filling and Replenishing of Gas

Nitrogen gas only may be used for filling or replenishing of the rubber bladders inside the accumulator and/or the piston accumulators. In connection with oil, oxygen or air may give rise to explosions!

4.2 Positioning of Valves (in case of N₂ Filling Pressure Checking and N₂ Pressure Increase)

In order to check both the hydraulic bladder accumulators and the hydraulic piston accumulators for **correct** nitrogen filling pressure, it will be absolutely necessary to completely relieve the pressure on the oil side. To this effect,

- the main switch should be set to the "Manual" position,
- and stop cocks should be opened according to the function table of the "Hydraulic Scheme HSLM".

In case of non-observance of this recommendation, an incorrect N₂ pressure will be indicated or gas will be replenished. In addition, the accumulator bladders may be destroyed.

4.3 Filling of Hydraulic Accumulators

Bladder Accumulators

For mill operation the rubber bladder will be filled with nitrogen from conventional steel bottles using a special filling device forming part of the scope of supply. Reading of filling pressure will be possible from a pressure gauge belonging to the filling device. In as-supplied condition, the accumulators will normally be filled with nitrogen to a sufficient pressure so that the rubber bladder will fill the entire steel jacket, i.e. will form a tight lining on the jacket and close the poppet valve on the oil side. In this way, the penetration of dirt into the hydraulic working area of accumulator will be prevented. There must be ensured that the gas valve will be sealed by means of the cap belonging to the accumulator. Closing of the gas valve alone will not ensure sufficient tightness.

Piston Accumulators

Filling of the upper piston accumulator chamber with nitrogen will be effected in a similar way. See also manufacturer documentation relating to the "Spring system, complete" assembly group of mill.

4.4 Technical Inspection

Hydraulic accumulators are pressure vessels and must be pressure tested at regular intervals by a competent inspection office according to the regulations valid in the country of installation. In most countries, the inspection of pressure accumulators and facilities operating with pressure accumulators will be mandatory and any such requirements must be strictly adhered to.

Hydraulic System

Mineral oils contain small amounts of entrained gas which can deaerate during mill operation.

The gas collects in large bubbles which can lead to damage to the hydraulic components unless it is regularly bled off (monthly). The quantity of gas varies with the oil type and operation of the mill and so the internals between bleeding must be adjusted to suit the practical situation.

Measurement connections at various positions within the hydraulic system serve as bleed points (see assembly drawing). The system must be bled until only oil without gas escapes from it.

Bladder Accumulators

After commissioning the gas charging pressure of the bladder accumulators must be regularly checked and, if necessary, corrected:

- after the first week;
- after the fourth week;
- after the third month;
- every half year.

Piston Accumulators

The piston accumulators can suffer on one hand from gas loss and on the other from oil transfer to the gas chamber caused by piston movements. This means that the following checks must be carried out at regular intervals:

Checking of piston accumulator gas-charging pressure:

- By a **pressure loss** of $\geq 30\%$ of the original charging pressure, the pressure is to be topped up again.
- By a **pressure gain** of $\geq 30\%$ of the original charging pressure there is a risk of oil being present in the gas chamber;

Procedure for checking:

- Using the filling-and-testing device relieve the gas pressure completely. Leave the valve screw open by 1.5 turns;
- Remove the filling-and-testing device;
- Mount a plastic hose (customer supply) over the gas valve body. Lead the free end to a container (customer supply);
- Initiate "Rollers lifting" at the hydraulic cabinet; when the pump begins to run it will move the piston of the accumulator upwards and expel the oil out of the gas chamber;
- Remove the hose, remount the filling-and-checking device and recharge the accumulator with gas (see chapter "4.3 Filling of Hydraulic Accumulators").

Inspection Intervals:

- after the first week;
- after the second week;
- after the fourth week;
- monthly;
- after the sixth month - every half year.

5. Protection of Concrete Foundation

Protect concrete foundations and any ducts therein from being destroyed by oil in that ducts for conveying oil and surfaces which could be wettened by oil, will be provided with a coating of oil-resistant paint.

6. Handwheel-operated High-Pressure Valves

When handwheel-operated high-pressure valves are used in the hydraulic cabinet (hydraulic cabinets) or in the connecting pipeline, attention should be paid that these valves will abut against the fixed stop when in open condition, in order to protect the stuffing box seal from excessive pressure and to prevent oil leakage at the valve stem.